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Capital flows under moral hazard

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ABSTRACT

I analyze a model with moral hazard and limited enforcement in a small open economy. I find that when state contingent contracting is allowed adding the moral hazard friction improves the model's predictions along several dimensions. First, it justifies why non-contingent debt is an optimal way to finance an emerging economy. Second, it explains the limited consumption risk-sharing and high, volatile and counter-cyclical interest rates. Third, it generates realistic crisis-like dynamics in which capital inflows are brought to a halt and interest rates sky-rocket. The model also has a strong internal propagation mechanism.

Limited enforcement friction, alone or together with moral hazard, has nearly no effect on the model's performance. I also construct a simple empirical test to distinguish between the two frictions and it favors moral hazard over limited enforcement friction.

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1. Introduction

With access to complete international credit markets, a typical economy should be able to diversify all of its country-specific risk. In states in which resources are scarce, the economy would borrow to finance a stable level of consumption and investment. However, emerging market economies seem to be unable to share idiosyncratic risk. First, this is reflected in the almost perfect correlation between consumption and output and consumption being at least as volatile as output. Second, emerging market economies face high, volatile and counter-cyclical interest rates. Third, emerging market economies often experience sudden stops: a decline in output that is accompanied by a sudden and sharp reduction of capital inflows. These three facts put emerging markets in stark contrast with small open developed economies like *e.g.* Canada. In this work I show how moral hazard that stems from lender's inability to monitor use of funds could rationalize the above facts. I also find that the model with moral hazard has a strong internal propagation mechanism and it generates realistic crisis-like dynamics. Presence of moral hazard is also supported by empirical evidence.

When markets are assumed to be exogenously incomplete, *e.g.* only non-contingent debt is traded, models with limited enforcement fit the data along several dimensions. Under limited enforcement a borrower cannot be forced to repay its debt; instead a country fulfils its obligations only if it is in its best interest. This line of research dates back to Eaton and Gersowitz (1981) that studies a deterministic economy subject to debt repudiation. Aguiar and Gopinath (2006, 2007) and Arellano (2008) extend the analysis in Eaton and Gersowitz (1981) to a stochastic environment. These models conform with the above three facts. Yet, they generate interest premia that are smaller and less volatile than in the data.[1] So, the literature concentrated on fixing the above deficiency by altering the after-default protocol, the only "free parameter" in their models. Arellano (2008) imposes a default penalty that varies positively with output at default. Yue (2010) added a possibility to negotiate hair-cut level in a Nash-bargaining game following default. Mendoza and Yue (2012) endogenize

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[1] Equivalently, there is too little borrowing in the model. If the default penalty is increased higher debt levels can be sustained in equilibrium but then interest premia fall.

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default penalty by positing that imported capital goods have to be replaced with imperfect domestic substitutes after default. In all these papers market incompleteness, in the sense of a limited set of financial instruments available to a borrower, is taken as given.

I start with complete markets that trade fully state-contingent debt, and ask what friction could simultaneously justify observed market incompleteness and conform with the three facts defining an emerging market economy. I compare limited enforcement and moral hazard frictions but, unlike in the previous studies, without restricting the type of contracts that can be written between a creditor and a borrower. Microfounding market incompleteness is important for policy analysis. Certainly, a policy prescription would be different depending on a source of market imperfections. If it is inability to enforce repayments, then building a strong country "credit history" could allow borrowing larger amounts in adverse states in the future. If it is moral hazard, then establishing a reliable and accurate system of national income accounts could give foreign investors more confidence. Deregulating a banking system could lead to more efficient and transparent resource allocation also mitigating the moral hazard problem. Establishing microfoundations for why markets are incomplete enables decision-makers to formulate relevant policy recommendations.

Unlike limited enforcement, the problem of moral hazard in international macroeconomics was given surprisingly little attention.[2] The data that is provided by the emerging market economies is often of low quality[3] opening possibilities for moral hazard. On the one hand, it is often difficult for creditors to sort out facts from fiction. On the other hand, it is easy for a borrowing nation to disguise consumption as investment or to "create" growth by misreporting main macroeconomic indicators. Also more frequently in emerging than in developed market economies investments are not channeled to their best uses for political reasons. As long as the allocation of funds is unobserved by creditors it leaves room for moral hazard.

Under moral hazard a country-borrower, after securing a loan, can choose low investment that is not consistent with amount of assumed debt. Creditors cannot verify country's investment and, thus, must structure repayments so that they induce borrower to take a desired action. This can be achieved by making repayments less responsive to economic conditions. While such strategy provides incentives for a borrower to invest it also limits insurance and burdens country with a larger debt. If the lowest output occurred repeatedly country's debt would grow quickly. In contrast, under limited enforcement country's debt would remain constant.

Gertler and Rogoff (1990) study a two-period model of international lending with moral hazard. In their model a risk-neutral agent borrows to finance a risky investment project. A creditor must provide incentives for investing by offering a spread (across states) in the borrower's net worth next period. As the authors point out, "[r]elative to the perfect-information benchmark, ... capital flows are dampened (and possibly reversed)." Investment is also lower relative to the frictionless economy and increases with the borrower's net worth, as observed in the data.

Atkeson (1991) extends the analysis in Gertler and Rogoff (1990) to a fully dynamic, infinite horizon setting with a risk-averse borrower. Contracting between the borrower and the lender is restricted both because the lender cannot observe the use of funds by the borrower (moral hazard) and the borrower can renege on the promised repayment (limited enforcement). Atkeson (1991) shows that a binding enforcement constraint may trigger capital outflows in the lowest output state.[4] In economies that are subject to the risk of repudiation, participation constraints usually bind in high output states. The presence of the moral hazard could reverse this result. Whether this can occur under a reasonable model parametrization is an open question.

In this work I revisit the model in Atkeson (1991). I prove that the first-order approach applies in this setting and solve the model numerically. I calibrate it to match Argentina's business cycle facts and find that the moral hazard friction severely limits the degree of state contingency built into the optimal contract. This result provides a justification for why non-contingent debt is an optimal way of financing an emerging economy. The moral hazard friction also generates high, volatile and counter-cyclical interest rates. The limited enforcement friction, on the other hand, adds little to the model's performance.

The moral hazard problem generates a strong internal propagation mechanism. The shocks that the economy faces are i.i.d. However, due to moral hazard investment often deviates significantly from the unconstrained optimal level. Large swings in investment lead to large and persistent deviations in output. A crisis scenario is 7.5 times more likely under the moral hazard friction than in the limited enforcement or frictionless economy. After a sequence of low output realizations the borrower gradually accumulates debt. Current account and debt increase gradually initially and then jump suddenly. Capital inflows stop and interest rates sky-rocket. Yet, interest rates react slowly to a worsening financial situation.

I also show how to extend the model to incorporate persistent output shocks. This improves the model's performance significantly. In particular, with persistent shocks consumption is nearly perfectly correlated with output and nearly as

[2] Researchers at the IMF studied the question whether its lending generates moral hazard. Rogoff (2002) gives a short overview of the arguments advanced and concludes that evidence on the presence of the moral hazard is mixed. The concept of moral hazard in this paper differs from that of the IMF.

[3] Economist (2011) writes about bogus inflation data published by INDEC, the statistical office of Argentina: "Since 2007 Argentina's government has published inflation figures that almost nobody believes. ... Misreported prices have cheated holders of inflation-linked bonds out of billions of dollars." This note suggests that governments are prone to tamper with official statistics to reduce its burden and to avoid bad headlines in difficult times.

[4] The condition for this result depends on the endogenous Lagrange multiplier and so cannot be verifies *a priori*. Under the parametrization assumed here this condition does not hold.

volatile. Finally, I argue that if the moral hazard friction is present correlation between consumption and output should be stronger conditional on low past output realization. This is consistent with the data.

Notably, the model parametrization does not rely heavily on differences in time preference between a creditor and a borrower. This means that an economy-borrower has a good chance to climb out of the debt. So, the model can explain both less successful emerging market economies in Latin America or Eastern Europe and more successful Asian economies.

The model is, however, unable to replicate current account data. In the data it is strongly counter-cyclical and very volatile while in the model it is neither. I explain that no exchange-economy model can simultaneously fit the observed consumption and current account data. I use a very stylized model, allowing state contingent contracting, to emphasize moral hazard that had not been investigated systematically before. This friction rationalizes use of non-contingent debt when lending to emerging market economies. It also allows the model to match several key data features. Therefore, I view the mechanism explained here as complementary rather than competing with the existing models of emerging market economies.

2. Empirical facts

In this work I analyze Argentina over the period from 1993:Q1 to 2005:Q4 as a representative emerging market economy. For comparison I choose Canada as a quintessential developed small economy.[5] Fig. 1 shows selected macroeconomic series for the two economies. Vertical dashed lines on each plot in the left column mark Argentina's default in December 2001. Around this period, output and private consumption plummeted (see panel a)—just in one year, from 2001:Q1 to 2001:Q4, real output per capita decreased by 11.6%. During the entire period, consumption closely followed output and declined by 13.3%, exemplifying Argentina's inability to diversify country risk.

Fact 1. *Emerging market economies are unable to diversify idiosyncratic risk.*

At the same time the EMBI[6] spread over the USD LIBOR, the premium at which Argentina could borrow, surged from 18 to 29% at the time of default and subsequently reached levels above 60%. These extraordinary spread levels effectively cut off Argentina from the international credit market. Interest rates in Argentina were also high. The spread between the rate on USD (fixed rate) loans to private non-financial institutions, as a measure of the cost of funds in Argentina, and USD LIBOR also increased significantly at the time of default and reached 33%. These loan spreads are also highly correlated with the EMBI country spread, as had been noted by Perri and Neumeyer (2005). For the period 1993:Q1–2001:Q4 correlation between these series is 0.96. In view of such a strong correlation I use bond spreads constructed by Perri and Neumeyer (2005) instead of unavailable loan spreads after 2001.

Fact 2. *Interest rates are strongly counter-cyclical.*

Up to the fourth quarter of 2001 Argentina had also been accumulating debt by running current account deficits. However, in 2001:Q4, the trade balance reversed to positive values. For the pre-default period, the correlation between detrended log real GDP/capita and the trade balance to GDP ratio is -0.81 (see Table 1). This means that Argentina on average experienced capital outflows (negative trade balance) when its output was high, and capital inflows when output was low.

Fact 3. *After a sequence of negative output realizations capital inflows either stop or reverse.*

The above stated facts are in stark contrast with observations on small open *developed* economies. Table 1 summarizes important data moments for Argentina and compares them with the corresponding moments for Canada. In the table, r denotes the spread between the domestic USD loan rate and the USD LIBOR.

Technology Shocks? Aguiar and Gopinath (2007) argue that volatile business cycles in many emerging market economies are caused by a volatile TFP trend. Garcia-Cicco et al. (2010) estimate a real business cycle (RBC) model allowing for both permanent and transitory productivity shocks using a long data sample for Argentina and Mexico. They find that an RBC model driven by technology shocks alone cannot explain business cycle statistics in the two emerging market economies.

It the beginning of the 2001 crisis in Argentina capital utilization was 48.2%. Assuming that the pre-crisis utilization level at 65% (observed in mid-2004), fluctuations in capital utilization alone can justify large swings in output.[7] This indicates that capital market imperfections, as assumed in my work, rather than volatile productivity is a cause of volatile business cycles in emerging market economies.

[5] Business cycle properties for a large set of emerging market economies are documented in Aguiar and Gopinath (2006) and Perri and Neumeyer (2005).

[6] This is the emerging markets' bond index computed by JP Morgan. It is considered one of the major indicators of a country's riskiness.

[7] Gertler et al. (2007) also find that capacity utilization, as measured by electricity use, can explain large movements in the measured productivity during the Korean crisis.

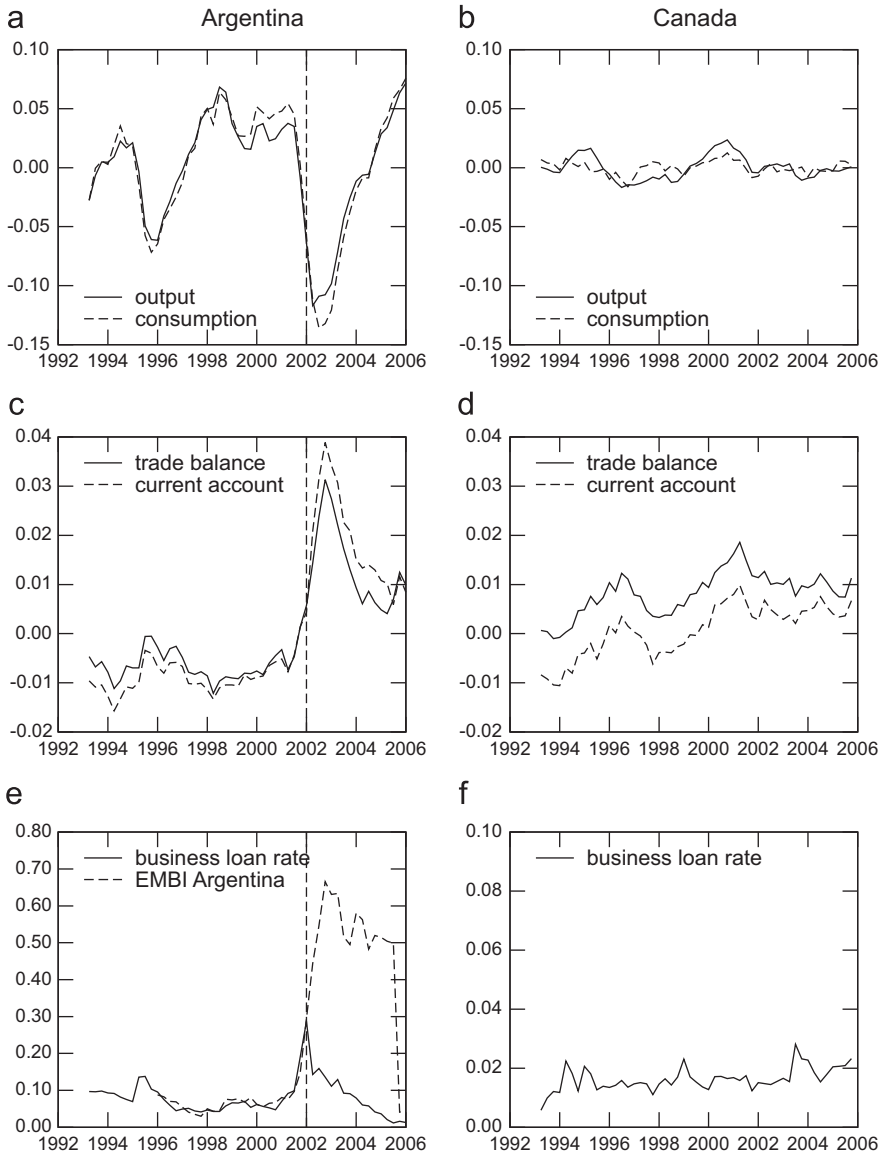

Fig. 1. Selected macroeconomic series for Argentina and Canada. Vertical dashed line marks Argentina's default. (a) Output and consumption, (b) output and consumption, (c) TB/GDP and CA/GDP, (d) TB/GDP and CA/GDP, (e) cost of borrowing—USD LIBOR and (f) cost of borrowing—CAD LIBOR.

Table 1
Data moments.[a,b]

Country	Period	$\frac{\sigma(c)}{\sigma(y)}$	$\rho(c,y)$	$\mu(r)$	$\sigma(r)$	$\rho(r,y)$	$\rho(tb,y)$	$\rho(tb,r)$
CAN	93:1–01:4	0.55	0.62	1.51	0.33	0.23	0.27	0.21
ARG	93:1–01:4	1.11	0.97	8.18	4.73	−0.58	−0.81	0.65
ARG	93:1–05:4	1.15	0.99	7.86	4.78	−0.68	−0.82	0.30
ARG	1980–2005	NA	0.95	11.00	4.89	−0.80	−0.77	0.89

[a] Output and consumption are filtered with Hodrick–Prescott filter.
[b] Output, consumption and trade balance-to-GDP ratio are seasonally adjusted.

3. Economic environment

Time is discrete and the planning horizon is infinite, $t \geq 0$. There is one perishable good at each date. There is one infinitely lived borrower. The borrower at each date receives a stochastic endowment and decides how to split available

recourses between consumption and investment. The borrower ranks alternative consumption streams according to

$$U^b = E_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \quad \beta \in (0,1), \quad u'(c) > 0, \quad u''(c) < 0. \tag{1}$$

Assumption A1 rules out allocations with zero consumption.

Assumption 1. The utility function u satisfies the Inada condition:

$$\lim_{c \downarrow 0} u'(c) = \infty.$$

The borrower starts period 0 with a given net worth n_0 which is a claim to n_0 units of date-0 good. Endowment at all consecutive dates depends on the previous period's investment, I. Having invested I units of date-t good, the borrower's endowment at date $t+1$ is drawn from a distribution $g(\cdot|I)$. It is assumed that, for all investment levels, the distribution of the next period's endowment has a fixed and finite support $Y = \{Y_1, \ldots, Y_s\}$ with $Y_1 < \cdots < Y_s$. Following Atkeson (1991), g is specified in the following way:

$$g(Y_j|I) = (1-\lambda(I))g_{0j} + \lambda(I)g_{1j}, \tag{2}$$

where $\lambda : R_+ \to [0,1]$ is an increasing and strictly concave function. Densities $g_0 > 0$ and $g_1 > 0$ are exogenously given and satisfy[8]:

Assumption 2. g_0 and g_1 satisfy monotone likelihood ratio condition.

The marginal effect of investment on the probability of state j occurring is

$$\frac{d}{dI} g(Y_j|I) = \lambda'(I)(g_{0j} - g_{1j}) \equiv \lambda'(I)\Delta g_j = \begin{cases} < 0, & i \text{ is small;} \\ \geq 0, & i \text{ is large.} \end{cases}$$

The international credit market is represented by a sequence of overlapping generations of lenders each living for two periods. A lender living in periods t and $t+1$ ranks alternative consumption streams according to

$$U_t^c = c_t + \beta_c E_t c_{t+1}, \quad \beta_c \in (0,1].$$

Lenders receive endowment, $M \gg 0$, in each period that they live and can borrow and lend at the risk-free rate of $1/\beta_c$. Presence of M emphasizes that the international credit market is not unlimited. Thus, the amount lent cannot exceed the lender's endowment:

$$b \leq M. \tag{3}$$

The borrower with net worth n faces the following budget constraint:

$$n + b = c + \theta I,$$

where b is the loan amount provided by a creditor, I is investment and $\theta > 0$ represents investment's cost.

Assumption 3. $\beta \leq \beta_c$.

This assumption reflects the fact that the government of an emerging economy may have a shorter planning horizon than a typical international lender.

Remark about production. The production process in this model is represented in a non-standard way. This is needed to facilitate the analysis. Yet, this specification includes the standard production function as a special case.[9]

3.1. Is investment unobservable?

A valid objection can be raised about observability of investment. I argue in the introduction that the NIA data in the emerging market economies is of poor quality. This leaves room for governments to distort macroeconomic statistics and anecdotal evidence points to this. Poor data quality could also be a consequence of different assumptions used by statistical agencies in emerging market and developed economies. Finally, sometimes it is difficult to draw a clear line

[8] Distributions g_0 and g_1 are said to satisfy monotone likelihood ratio property if g_{0j}/g_{1j} is increasing in j.

[9] First, note that it is a concave production function in the sense that expected output is a concave function of investment:

$$\frac{d^2}{dI^2}\left(\sum_{j=1}^{s} g_j(I)Y_j\right) = \lambda''(I)\sum_{j=1}^{s} \Delta g_j Y_j < 0.$$

Next, consider the example in which log of output $y_j = log(Y_j)$ is distributed according to $(1-\lambda(I))\Phi(y_j|0,1) + \lambda(I)\Phi(y_j|1,1) = \Phi(y_j|\lambda(I),1)$, where $\Phi(\cdot|\mu,\sigma)$ denotes normal distribution with mean μ and standard deviation σ. Assuming $\lambda(I) = \alpha \cdot \log(I)$, the output process can be also represented as $Y_j = z_j I^{\alpha}$ where $\log(z_j)$ is a standard normal random variable. In view of the above current specification includes as a special case the model in Clementi and Hopenhayn (2006) albeit with one-period capital.

between investment and consumption. Is an oversized presidential jet or an ad campaign of a bridge that is never built consumption or investment?

Another possibility is a world in which investment level, but not its quality or its productivity, is observable.[10] Productivity of investment could depend on such hard-to-measure factors as rule of law enforcement, bureaucracy and corruption levels. Another factor could be misallocation induced by subsidies and restrictions on trade. For example, in 2008 Argentina voted that cell phones must be produced domestically and awarded a subsidy to a foreign producer. In the same year Argentina's farmers were left with a significant amount[11] of maize that could be neither sold in the satiated domestic market nor exported abroad due to export quotas. Earlier quotas on beef exports, aimed to shift production out of agriculture and into manufacturing, drove producers out of business. To offset some of the costs stemming from export quotas the government started to offer loans to exporters at a negative interest rate.[12] These incidents point to a massive misallocation of resources and suggest that "effective" investment is indeed difficult to measure. No doubts, this welcomes moral hazard.

In a related work Clementi et al. (2010) interpret I as effort.[13] But it seems that poor data quality and corrupted resource allocation are a more likely explanation of why the "effective" investment may not be observable.

3.2. Contracts and frictions

Each period, the borrower signs a contract $\mathcal{C}$ with the creditor. In a frictionless economy (in which the borrower's investment is observed and contracts are enforceable) $\mathcal{C}$ specifies (1) the loan amount b provided by a creditor, (2) the repayment schedule $(d_1,..,d_n)$, where d_j is the amount to be repaid by the borrower in state j, and (3) the investment, I, made by the borrower. I analyze the unconstrained efficient contract in Section 3.3.1.

The first friction that I study is *moral hazard*. Output is observable which means that repayment can be made contingent on output realization. However, the lender is able to observe neither investment nor consumption. This induces a moral hazard problem. A lender can no longer fully insure the borrower as the latter would optimally invest nothing. Formally, moral hazard requires investment to be part of the borrower's optimal response to a contract (b,d), rather than being specified explicitly in a contract.

The second friction that I study is *limited enforcement*. It is assumed that contracts with a sovereign nation cannot be enforced. If a borrower decides to renege on a contract, he is forever excluded from the international credit market. A creditor has to design a contract so that the borrower always (weakly) prefers to participate in the international credit market. Formally, a limited enforcement restriction imposes an upper bound on repayments that a creditor can demand from the borrower.

In what follows I study recursive economies. That is an optimal contract and the borrower's investment decision are functions of the borrower's current net worth. For the economy with moral hazard and enforcement constraints the existence of a recursive representation is shown by Atkeson (1991) in proposition 5 (page 1083). The existence of a recursive representation for the economy with only one of the frictions follows directly. I also take for granted that the optimal value functions are strictly increasing and strictly concave.

3.3. Two benchmarks

3.3.1. Frictionless credit market

In this section I analyze the case when a borrower has access to a frictionless credit market. The borrower's investment can be contracted upon. There is no enforcement problem on either side of the market. The value function defined in this section is an upper bound on the utility that can be achieved in the environment with moral hazard and limited enforcement studied later.[14]

Because it is a frictionless environment I can study the equivalent economy in which a creditor and the borrower trade a full set of Arrow securities contingent on the borrower's performance. Arrow security j pays one unit of good if state j occurs tomorrow. Because the lender is risk-neutral, Arrow securities are traded at

$$q(Y_j|I)=\beta g(Y_j|I). \tag{4}$$

Let d_j be the quantity of Arrow security j issued by the borrower. Let $V_{\mathrm{ad}}(n)$ be the optimal value to the borrower with net worth n. The value function $V_{\mathrm{ad}}(n)$ satisfies the following Bellman equation:

$$V_{\mathrm{ad}}(n)=\max_{I,b,d}\left\{u(n+b-\theta I)+\beta\sum_{j=1}^{s}g(Y_j|I)V_{\mathrm{ad}}(Y_j-d_j)\right\}, \tag{5}$$

[10] I thank Gianluca Violante for this suggestion.

[11] Based on my estimates about 25% of that year's annual production was affected.

[12] These facts on Argentina were taken from Economist (2011).

[13] In their work the one period utility is separable in consumption and effort: $u(c_t)-\theta I_t$. This specification, while simplifying the analysis considerably, implies that wealthier economies counter-factually invest less.

[14] For a numerical solution it is important to start iterations from the largest possible value function. Otherwise iterations may converge to the value of financial autarky defined in the next section.

subject to the lender's individual rationality condition[15]:

$$b \le \beta_c \sum_{j=1}^{s} g(Y_j|I)d_j, \tag{6}$$

where b is the amount of good that the borrower receives in exchange for his promise, $\{d_j, \forall j\}$, to deliver goods tomorrow. Loan amount b must also satisfy Eq. (3): $b \le M$. If this constraint were absent then a borrower would always invest the optimal amount I^* that satisfies the following relation[16]:

$$\theta = \beta\lambda'(I^*) \sum_{j=1}^{s} \Delta g_j Y_j. \tag{7}$$

Let E^* denote the expectation with respect to the distribution $g(Y_j|I^*)$. Denote the lowest equilibrium net worth level by $n_c \equiv E^*(Y_j) - M/\beta_c$. The optimal repayment schedule delivers state non-contingent net worth n' next period: $d_j = Y_j - n'$, $\forall j \in S$. When $n \le n_c$ then net worth is increasing: $n \le n'$. In this case the borrower is financially constrained: $b = M$. When $n > n_c$ then net worth is decreasing: $n > n'$. The borrower may or may not be financially constrained. Irrespectively of its initial level the borrower's net worth converges to n_c. Investment and consumption being continuous functions of net worth also converge to a constant level.

Current account (capital outflow if positive) each period is

$$ca_t = ca(n_t) \equiv d(Y_t, n_{t-1}) - b(n_t) = Y_t - n_t - b(n_t).$$

Since transition from n_{t-1} to n_t does not depend on output realization Y_t, the correlation between current account and output equals 1. Consumption is eventually constant and, so, uncorrelated with output. The frictionless model implies uninhibited risk-sharing and a strongly pro-cyclical current account.

3.3.2. Financial autarky

Assume that the borrower has no access to the international credit market (financial autarky) and the investment technology is the only vehicle for inter-temporal consumption smoothing. Let $V_{\text{aut}}(n)$ be the optimal value to a borrower with $n \ge 0$ units of good living in autarky. This value function is the lower bound on the values achieved in other environments. It satisfies the following Bellman equation:

$$V_{\text{aut}}(n) = \max_{I \in [0,n]} \left\{ u(n - \theta I) + \beta \sum_{j=1}^{s} g(Y_j|I) V_{\text{aut}}(Y_j) \right\}. \tag{8}$$

The optimal investment made by the agent is an increasing function of n.

Next consider the environment in which the borrower upon entering a financial autarky loses a fraction $(1-\delta) \in [0,1)$ of output. Parameter δ governs "perceived" cost of default.[17] The optimal value to the borrower then is

$$V^{\delta}_{\text{aut}}(Y_j) = \max_{I \in [0,Y_j]} u(\delta Y_j - \theta I) + \beta \sum_{k=1}^{s} V_{\text{aut}}(Y_k). \tag{9}$$

In what follows I assume that the borrower entering the autarkic regime must suffer an output loss.

4. Moral hazard and limited enforcement

In this section I assume that contracting between the borrower and the lender is restricted both because the lender cannot observe the use of funds by the borrower (moral hazard) and because the borrower can renege on the promised repayment (limited enforcement). Thus, a creditor cannot perfectly insure the borrower as then no investment will be made. In addition, a creditor has a limited choice of repayment schedules as a large repayment may induce a default.

Atkeson (1991) shows that in this environment the optimal allocation can be implemented by a sequence of one-period contracts that depend on the borrower's net worth. Let $V(n)$ be the optimal value to the borrower with net worth n under the optimal contract. The value function $V(n)$ satisfies the following Bellman equation:

$$V(n) = \max_{b,d} \left\{ u(n + b - \theta I) + \beta \sum_{j=1}^{s} g(Y_j|I) V(Y_j - d_j) \right\}, \tag{10}$$

subject to

$$b \le \beta_c \sum_{j=1}^{s} g(Y_j|I) d_j, \tag{11}$$

[15] According to this condition, the lender's gross rate of return is no less than $1/\beta$.

[16] I^* is also the level of investment that would be chosen by the lender if he were allowed to operate the investment technology directly.

[17] We refer to $1-\delta$ as perceived rather than actual cost of default because the borrower never chooses to default under the optimal contract.

$$I \in \arg\max_{\hat{I}\in[0,n+b]} \left\{ u(n+b-\theta\hat{I}) + \beta \sum_{j=1}^{s} g(Y_j|\hat{I}) V(Y_j - d_j) \right\}, \tag{12}$$

$$V^{\delta}_{\text{aut}}(Y_j) \le V(Y_j - d_j), \quad \forall j \in S. \tag{13}$$

According to the constraint (11) the loan amount cannot exceed the present discounted value of the repayments. The constraint (12) is the borrower's incentives compatibility constraint. It states that given the contract (b,d) the borrower chooses investment optimally. The constraints (13) are the enforcement constraints. They state that the borrower in every state weakly prefers honoring his debt obligations to exclusion from the credit market (autarky).

4.1. Relaxed first-order approach

Constraint (12) is not tractable as it is. Replacing it with the first-order condition is a viable alternative. But, first, it has to be shown that the first-order approach is valid in this setting.[18] Following Rogerson (1985) I replace constraint (12) with the following inequality:

$$-\theta u'(n+b-\theta I) + \beta\lambda'(I) \sum_{j=1}^{s} \Delta g_j V(Y_j - d_j) \ge 0. \tag{14}$$

I need to show that replacing (12) with (14) in the maximization problem (10) does not affect the solution. For this I need to prove that for any feasible contract (b,d) the following two conditions are satisfied:

$$-\theta u'(n+b-\theta I) + \beta\lambda'(I) \sum_{j=1}^{s} \Delta g_j V(Y_j - d_j) = 0,$$

$$\theta^2 u''(n+b-\theta I) + \beta\lambda''(I) \sum_{j=1}^{s} \Delta g_j V(Y_j - d_j) < 0.$$

The second inequality implies that the borrower's objective is strictly concave in investment. Given that u and λ are strictly concave a sufficient condition for the second inequality is $\sum_{j=1}^{s} \Delta g_j V(Y_j - d_j) \ge 0$.

Lemma 1. *Replacing constraint (12) with the relaxed first-order condition (14) does not change the solution to the problem (10).*

Proof. Fix a contract (b,d). The corner $I = (n+b)/\theta$ is not a solution because of assumption 1.

Suppose that $\sum_{j=1}^{s} \Delta g_j V(Y_j - d_j) < 0$. Then optimal investment must be zero. Also there must exist $j < k$ such that $Y_j - d_j > Y_k - d_k$. But then the enforcement constraint in state j cannot be binding. Without loss of generality, assume that $(j,k) = (1,2)$ and consider the following alternative contract: $\tilde{b} = b, \tilde{d}_1 = d_1 + e/g_{01}$, $\tilde{d}_2 = d_2 - e/g_{02}$, $\tilde{d}_3 = d_3, \ldots, \tilde{d}_s = d_s$ for some $e > 0$. It is feasible and it implies zero investment if e is sufficiently small. Since V is strictly concave we have $\sum_{j=1}^{s} g_j(0) V(Y_j - d_j) < \sum_{j=1}^{s} g_j(0) V(Y_j - \tilde{d}_j)$: the alternative contract implies a strictly better continuation value for the borrower. So, the original contract could not be optimal and we must have: $\sum_{j=1}^{s} \Delta g_j V(Y_j - d_j) \ge 0$. Then the borrower's objective must be strictly concave because $u'', \lambda'' < 0$. In this case the relaxed first-order condition (14) must hold with equality.[19] □

4.2. Properties of optimal contract

Assume temporarily that a creditor has unlimited resources: $M = \infty$. As shown in Appendix A, the optimal contract must satisfy the following optimality condition

$$V'(n) = \frac{\beta}{\beta_c} V'(Y_j - d_j) \left(1 + \mu \frac{\lambda'(I)\Delta g_j}{g(Y_j|I)} + \gamma_j \right), \tag{15}$$

where $\mu \ge 0$ is the Lagrange multiplier associated with the borrower's incentives constraint and $\gamma_j \ge 0$ is the scaled Lagrange multiplier associated with the limited enforcement constraint for state j.

I now explain how different frictions affect the economy's dynamics. I first assume that $\beta = \beta_c$. Under perfect contract enforcement, $\gamma_j = 0, \forall j$, the Euler equation (15) implies the following relation:

$$V'(n) \le E[V'(Y_j - d_j)]. \tag{16}$$

So, $V'(n)$ is a sub-martingale. Given concavity of the value function, this means that moral hazard problem forces the borrower's net worth to decline on average. Yet, next period net worth $n_j' = Y_j - d_j$ increases when $\Delta g_j > 0$ and declines otherwise. This is how a creditor provides incentives for investment: by spreading the continuation value of the borrower across future states a creditor unloads part of the risk to the borrower. Incentives provision is costly—it reduces expected

[18] Note that applicability of the first-order approach also simplifies computation of the optimal contract.

[19] This lemma proves indirectly that Assumption 6 in Atkeson (1991, p. 1085), necessary for validity of the first-order approach, must hold.

continuation value. The latter must be compensated by an increase in current consumption. Thus, moral hazard problem leads to excessive consumption and immiseration that is also present in the private information economies of Thomas and Worrall (1990) and Atkeson and Lucas (1992) among others.

When investment is contractible but there is limited enforcement, $\mu = 0$, the Euler equation (15) implies the following relation:

$$V'(n) \geq E[V'(Y_j - d_j)]. \tag{17}$$

That is the borrower's net worth is expected to drift up on average. When the lowest output realization occurs the borrower's net worth remains the same. The repayment in the low output state is small and there is a capital inflow not outflow as observed in the data. That is the effect of limited enforcement is the opposite to that of moral hazard. Finally, the borrower's net worth eventually converges to a level at which enforcement constraints are non-operative. That is the limited enforcement economy is "in the long run" indistinguishable from the frictionless economy.

Differential discounting $\beta < \beta_c$ encourages benefit front-loading. That is a creditor rewards the borrower with current consumption rather than by promising high utility/consumption in the future. This effect is similar to that of moral hazard. Even small differences in discounting significantly speed up convergence to a stationary distribution of net worth. Yet, because of moral hazard and Eq. (16) existence of a stationary distribution of net worth is guaranteed even with equal discounting.

Limited enforcement imposes state contingent limits on next period's net worth. Because V is strictly increasing, Eq. (13) can be rewritten as

$$d_j \leq \overline{d}_j \equiv Y_j - V^{-1}(V^{\delta}_{\text{aut}}(Y_j)). \tag{18}$$

Because $V(n) \geq V^{\delta}_{\text{aut}}(n) \quad \forall n$ we have $\overline{d}_j \geq 0$. In what follows I refer to the restriction (18) as the endogenous borrowing limit.[20]

Fig. 2 plots the Lagrange multipliers for the parameter values assumed in Table 2. The Lagrange multiplier μ on the borrower's incentives compatibility constraint (12) measures the strength of the moral hazard problem. It is shown in panel A. It is decreasing on the set of net worth levels where the multipliers on the endogenous borrowing limits, shown in panel B, are zero. When the borrower is wealthy and can afford higher investment the cost of incentives provision is low. As net worth declines moral hazard problem becomes more severe. But when the borrowing limits bind the strength of the moral hazard problem declines abruptly as net worth decreases. It collapses to zero where the borrowing limits bind in each state of the world. The reason is that the limited enforcement alone pressures a creditor to spread the continuation values beyond what is needed to mitigate the moral hazard problem.

Note that the Lagrange multiplier on the endogenous borrowing constraint for state 2 is positive on a larger set than the multiplier on the constraint for state 1. A creditor would rather increase payment in the high output rather than the low output state. That is, like in the model with limited enforcement alone, the borrowing limit for the high output state is most restrictive. Moral hazard, unfortunately, does not reverse this pattern.

Atkeson (1991) shows that if moral hazard is sufficiently strong (μ is large) then after a sequence of low-output realizations the economy should reach the borrowing limit and there should be capital outflows. This result does not extend to the current environment because of the default output penalty.[21] But the current solution still can be used to demonstrate the forces that could lead to the result desired by Atkeson. The idea is that if μ is sufficiently large and $1 + \mu\lambda'(I)\Delta g_1/g_1(I) < 0$ then by (15) we must have $\gamma_1 > 0$. That is if moral hazard problem is sufficiently severe a creditor requires the maximum repayment after a low output realization. This triggers capital outflows *if* the repayment in the high output state is less than the maximum.[22] Yet, Fig. 2 shows that this is not the case. Moreover, the Lagrange multiplier μ vanishes where the two borrowing limits bind simultaneously. That is limited enforcement "turns off" moral hazard instead of interacting with it.

5. Quantitative analysis

5.1. Parameterization

I assume that one period corresponds to a quarter. The one period utility function is $u(c) = c^{1-\gamma}/(1-\gamma)$ with $\gamma = 2$. The discount factor of a creditor β_c is set to 0.990. I choose the borrower's discount factor to be 0.980 that is very close to the discount factor of a creditor. Even a small difference in discounting significantly reduces the equilibrium support of the borrower's net worth. The difference that is assumed here is small relative to what is assumed in other studies cited in this paper. This is significant because when the creditor's discount factor is much higher than that of the borrower the latter spends most of the time near the borrowing limit. That is the model is set so that the borrower can never reduce the debt to a "manageable" level, even in the unlikely event that the highest output realizes ad infinitum.

[20] Alvarez and Jermann (2001) refer to the same restriction as the endogenous solvency constraint.

[21] If output loss were permanent the result in Atkeson (1991) would apply again. But such an assumption seems implausible. When $\delta = 0$ and other parameters are unchanged there are no capital outflows.

[22] Intuitively, when repayment in the high output state is less than the maximum the loan amount (capital inflow) is not as large as it could be.

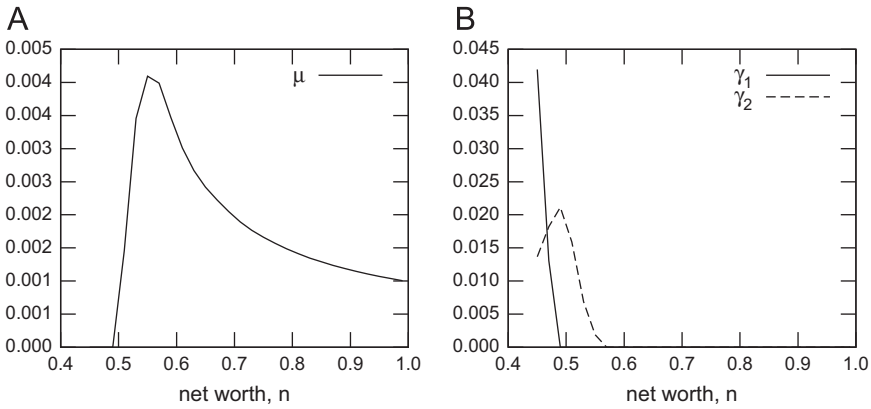

Fig. 2. Lagrange multipliers in the model with limited enforcement and moral hazard. (A) Incentives constraint and (B) endogenous borrowing limits.

Table 2
Model parameters.

Parameter	Value	Data moment
β_c	0.990	World interest rate=0.040
M	0.465	Average debt/output ratio=0.410
$\ln(Y_j)$	$\pm$ 0.054	Output volatility=0.054
θ	0.105	Average output level=1.000
δ	0.795	Average debt/output ratio=0.420
γ	2.000	–
β	0.980	–
ν	0.950	–

Output process is governed by parameters θ,λ,g_0,g_1. I assume only two states and the following distribution:

$$prob(Y_1|I)=1-\lambda(I), \quad prob(Y_2|I)=\lambda(I). \tag{19}$$

The closer is λ to a linear function the more volatile is investment.[23] So, ideally λ would be a linear function. However, in this case optimal investment can be a corner solution invalidating the first-order approach. So, I assume the following functional form[24]:

$$\lambda(I)=\min(I^{0.95},1). \tag{20}$$

Parameter θ is set to match the desired expected level of output that I normalize to 1. States Y_1 and Y_2 are chosen to match the desired level of income volatility. Endogeneity of the income process requires using the simulated method of moments to compute model parameters.

Parameters δ and M are set to match the average debt to output ratio respectively in the full model and the model with moral hazard only.[25] The assigned values are 0.795 and 0.465, respectively. The value of δ is low but not unreasonable.[26] All the assumed parameter values are gathered in Table 2.

5.2. Model solution

I compute the model solution by solving the Bellman equation iteratively. Value function is approximated by a cubic spline on the interval $\mathcal{N}\equiv[0.2,1.2]$ with 100 nodes. Stopping criterion is: $\sup_{n\in\mathcal{N}}|V^{k+1}(n)-V^k(n)|<10^{-5}$. Further computational details are provided in Appendix B.

The solutions to the model with moral hazard only ("MH") and limited enforcement only ("LE") are presented in Fig. 3. I first describe the optimal policies in the MH model and later compare them to those in the LE model. The solution to the MH+LE model is not discussed because it is visually identical to the MH solution. The vertical dashed and dotted lines in all panels mark the lowest level of net worth in the MH and the LE models respectively.

[23] The more volatile is investment, the more model deviates from the frictionless setting.

[24] I also tried to model $\lambda(I)$ by a fourth-order polynomial. The best fit turns out to be numerically close to the one presented here.

[25] In the model with moral hazard I impose the constraint $b\le M$ while in the full and the limited enforcement models I drop this constraint.

[26] I need such a large output penalty because autarkic values are high due to self-insurance. Self-insurance via investment is not available in pure exchange economies. Thus, in the latter it is possible to match the observed debt level with a much lower output penalty. This penalty, however, except for changing the average debt level has virtually zero effect on model performance as demonstrated in the online appendix.

The solid dark line in panel A represents the optimal weight on the high output outcome, $\lambda(I_{\mathrm{MH}}(n))$. At high levels of net worth investment is more affordable and equals the unconstrained optimal level of investment I^*. At low levels of net worth investment is sensitive to changes in the financial position of the borrower. The weight on the high output outcome fluctuates between 0.20 and 0.60. The portion of the investment policy function with a positive slope is responsible for generating persistence. After a low output realization net worth declines, investment declines and probability weight on low output outcome increases.

Panel B plots expected future net worth, $E[Y_j - d_j(n)]$. At high levels of net worth the slope of the LE line is $\beta/\beta_c < 1$. The slope of the MH line is smaller by 0.005. This means that net worth drifts down faster in the MH economy. While the difference is visually unnoticeable it is equivalent to increasing the borrower's discount rate by 2% per annum. At low levels of net worth $E[Y_j - d_j(n)]$ is decreasing which means that endogenous improvement in output distribution (implying a higher probability of the large repayment) dominates the effect of increasing net worth.

Panel C plots optimal loan $b_{\mathrm{MH}}(n)$ and repayment schedule $\{d_{1,\mathrm{MH}}(n), d_{2,\mathrm{MH}}(n)\}$. All three lines are extremely close: nearly all the risk is assumed by the risk-averse borrower. That is the insurance comes in the form of access to borrowing. At low levels of net worth the loan and repayment schedules are constant.

The dark lines in panel E represent optimal current account $ca_j(n), j \in \{1,2\}$. Current account depends on current output—it determines the repayment made in the current (but agreed to in the previous) period. The distance between the two lines is $Y_2 - Y_1$. When the borrower is relatively poor (wealthy) current account reacts strongly (weakly) to changes in net worth. Suppose that output changes from Y_2 to Y_1. Because of insurance motives the direct effect of the output decline is to decrease current account (reduce capital outflow). However, incentives provision requires a decrease in the borrower's net worth. The indirect effect of the output decline, thus, is to increase current account (increase capital outflow). At high levels of net worth the insurance effect dominates; hence, current account and output are positively correlated. At low levels of net worth the incentives effect dominates and $\rho(ca_t, y_t)$ becomes smaller or even negative. Whether current account is counter-cyclical or not depends on what fraction of time the economy spends in the low net worth region, roughly below 0.56. Under the current parametrization, it spends 51% of the time in the "high sensitivity" region where investment is an increasing function of the borrower's net worth.[27] Fig. 1 in the online appendix plots histogram of the country's net worth.

I now turn to the LE economy. Optimal weight on high output outcome, light solid line in Fig. 3 panel A, is much higher than in the MH economy. Investment is high because it is observable and a creditor rationally dictates to invest more than a borrower would. But it is also less volatile. For this reason, the LE economy's endogenous propagation mechanism is weak. Current account, light solid and dashed lines in Fig. 3 panel E, also vary less with the borrower's net worth. This means that it is nearly impossible for current account to be counter-cyclical in the LE economy.

Fig. 3 panel D plots the components of the optimal contract for the LE economy. Compare it with the MH contract in Fig. 3 panel B. Most notably the LE contract offers a significant amount of insurance. To quantify the degree of insurance provided by different contracts I compute the following "risk-sharing index":

$$RSI(n) = \frac{d_2(n) - d_1(n)}{Y_2 - Y_1}. \tag{21}$$

In the frictionless economy this index equals one, its maximal value. This index equals zero, its minimal value, when the repayment is state non-contingent: $d_2(n) = d_1(n)$. Fig. 3 panel F plots this index for the LE and the MH economies. When the moral hazard is present repayment is essentially state non-contingent: the maximal value of RSI in the MH model is below 0.01. In contrast, in the limited enforcement economy this index is on average 0.80.[28]

5.3. Interest rate spread

One of the main objects of interest in this work is an interest rate spread. I propose the following reinterpretation of the current setup. Consider a government-borrower that instead of a contract faces an interest rate schedule $R(n)$. That is the government of the economy with net worth n promises to repay $R(n)$ for each unit borrowed. The government "taxes away" the excess return (capital outflows) from the risk-neutral creditors to finance future consumption and stimulate investment. The implied interest rate schedule is

$$R(n) = \frac{u'(c(n))}{\beta \sum_j g_j(I(n)) u'(c(Y_j - d_j(n)))}, \tag{22}$$

where $I(n)$, $d_j(n)$, $c(n)$ are the policy functions under the optimal contract. Fig. 2 in the online appendix shows the implied (annualized) interest rate schedule $[R(n)]^4$. For high levels of net worth the implied interest rate coincides with the world level. Interest rate spread increases significantly at low levels of net worth reaching a maximum of 12% above the world level.

[27] By assuming a larger discount factor differential it is possible to force the economy completely into the "high-sensitivity" region. However, such "behavioral" parameterizations are not realistic.

[28] Yet, it varies in equilibrium between -0.09 and 1.0.

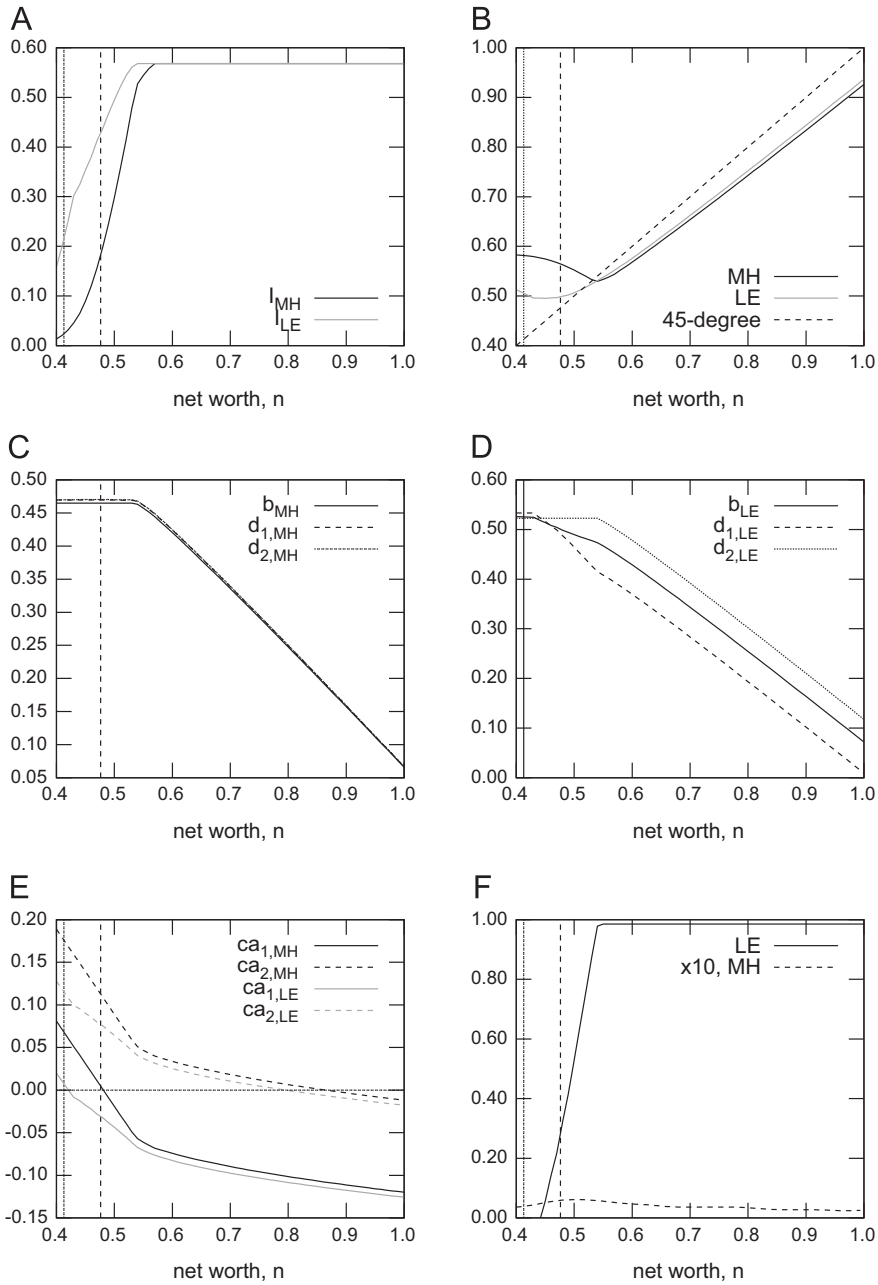

Fig. 3. Optimal policies in the MH and the LE economies. Vertical dashed and dotted lines denote the lowest net worth level respectively in the MH and the LE economies. (A) Investment, (B) expected future net worth, $E[Y_j - d_j(n)]$, (C) MH loan and repayment, (D) LE loan and repayment, (E) capital outflows and (F) risk-sharing index.

5.4. Simulation results

First, I describe how the economy evolves in a crisis-like situation. Fig. 4 shows the dynamics of the MH and LE economies on a path of low output realizations. Consider first the MH economy. Panel A shows that starting with zero debt the economy steadily accumulates obligations. After four quarters debt/output ratio reaches about 0.4. After five quarters the economy nearly exhausts its borrowing capacity. That is when the interest rate jumps to 14% as shown in panel B. Panel C shows current account that first increases gradually (that is capital inflows gradually shrink) and then in quarter five it jumps to zero and later increases slightly. In this economy the interest rate gives a late warning about the economy's health, unlike debt level and current account. While such the path of output is unlikely (it's probability is only 0.0024) it is

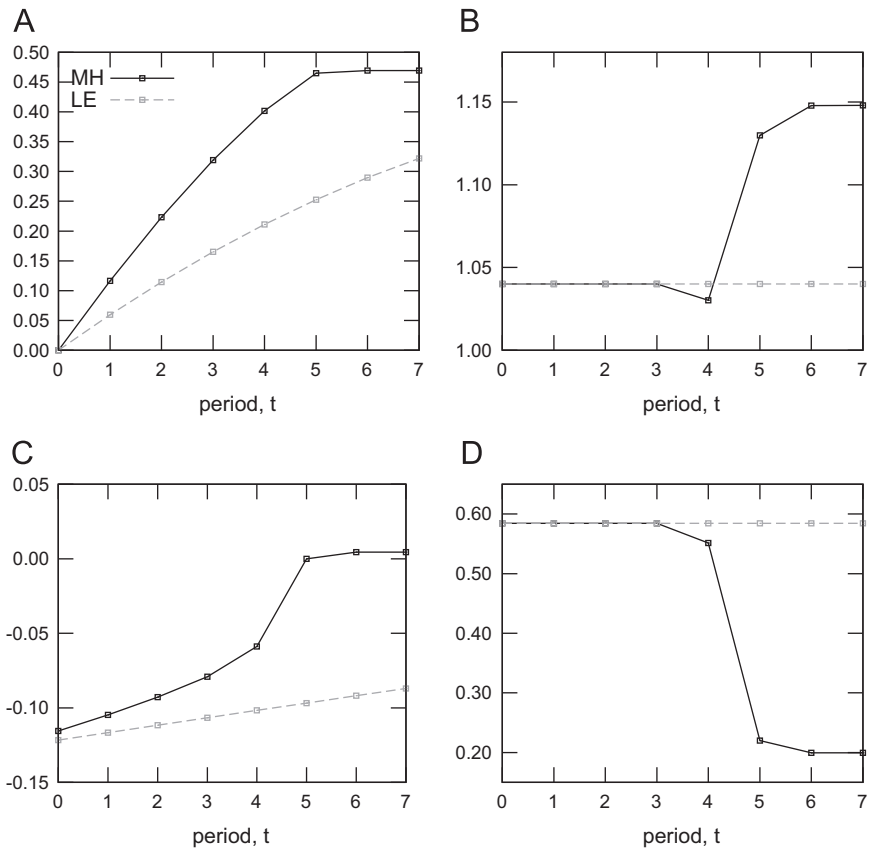

Fig. 4. A crisis scenario in the MH and the LE economies. (A) Debt/output, (B) interest rate, (C) current account/output and (D) Prob(Y_2).

7.55 times more likely to occur in this model,[29] that is under moral hazard, than in the model without frictions. The dynamics of the LE economy is trivial because it takes another four periods until the country's borrowing rate deviates from the world level.[30]

I now turn to the model moments that are reported in Table 3.[31] These results were obtained by simulating time series of length 20,000 starting from n_0 that is the mid-point of the net worth' support. The first half of the sample was discarded. Note that I report unconditional moments while other studies report moments conditional on a sample path leading to default. In what follows I describe the MH results, but my discussion also applies verbatim to the MH+LE model. I use the LE model as a comparison benchmark.[32]

First, the model reproduces exactly moments of the output process and the average debt/output ratio. Both are endogenous. While output persistence stands at 0.28 it is worth noting that the shocks in the model are i.i.d. That is the model has a strong propagation mechanism.[33] Second, interest rate spread in the model matches the data well. It is substantial, 3.84%, and it is very volatile, 5.20%. The spread is also strongly counter-cyclical. After a bad realization of output net worth declines. The economy moves closer to the borrowing limit and interest rate increases giving the borrower incentives to invest. Third, consumption volatility and its correlation with output are significantly increased relative to the LE model. These three results are a consequence of "benefit front-loading:" the optimal MH contract forces the borrower toward low net worth level over time. So, the economy spends more time in the region where its borrowing is constrained and the limited ability to smooth consumption is reduced further. Interest rates facing the borrower are increased to prevent it from accumulating even more debt. The optimal LE contract forces the country toward high net worth levels. Relative impatience of the borrower is the only force that makes it borrow. So, while the interest rates can be high such periods are infrequent. Most of the time the borrower is charged no premium and this makes interest rate spread less counter-cyclical than under the MH contract.

[29] The likelihood of the crisis path is given by $prob(Y_1|n_8 = Y_1 - d_1(n_7)) \ldots prob(Y_1|n_1 = Y_1 - d_1(n_0))prob(Y_1|n_0 = Y_1)$.

[30] To put it differently, even if the LE economy was started with half of its average debt its borrowing rate would still equal the world rate after eight periods.

[31] Because quarterly series for investment for Argentina are not available I used 1993–2005 annual series from Garcia-Cicco et al. (2010).

[32] This model turns out to be very similar to the frictionless model which is not reported here due to space constraints.

[33] For example, a workhorse RBC model offers no internal propagation.

Table 3
Model moments.

Moment	Data	i.i.d. shocks			Persistent shocks		
		MH+LE	MH	LE	MH+LE	MH	LE
$E(r)$	8.18	3.90	3.84	4.27	3.19	3.26	4.01
$\sigma(r)$	4.73	5.71	5.20	2.51	5.52	4.73	1.96
$\sigma(c)/\sigma(y)$	1.11	0.72	0.72	0.24	0.84	0.84	0.40
Cross-correlations							
$\rho(c,y)$	0.97	0.82	0.82	0.62	0.96	0.96	0.67
$\rho(r,y)$	−0.58	−0.65	−0.68	−0.81	−0.65	−0.74	−0.42
$\rho(tb,y)$	−0.81	0.70	0.69	0.98	0.62	0.62	0.92
Investment							
$\sigma(i)/\sigma(y)$	2.96	8.28	9.11	2.29	6.69	7.87	3.18
$\rho(i,y)$	0.95	0.76	0.76	0.46	0.97	0.95	0.58
Income and debt							
$\rho(y)$	0.94	0.27	0.28	0.07	0.75	0.75	0.63
$\sigma(y)$	5.40	5.40	5.40	5.40	5.40	5.40	5.40
$\mu(d/Y)$	0.42	0.42	0.42	0.51	0.42	0.42	0.34
Crisis likelihood		7.31	7.55	1.01	3.01	3.25	0.99

But the MH model's success ends here. Consumption volatility is significantly smaller than in the data. It is also less correlated with output than in the data. That is the MH model still allows "too much" insurance. As a consequence current account is not sufficiently volatile and is pro- rather than counter-cyclical. This failure is, however, expected.[34]

Finally, I would like to note that investment in the MH economy is about three times more volatile and it is not as correlated with output as it is in the data. The reason for the first outcome is that investment is less than 5% of output; so, its extreme volatility imposes a minimal cost on the borrower. The reason for the second outcome is that the economy spends a substantial amount of time in the region where investment is stable, see Fig. 3 panel A. This can actually be viewed as a virtue of the MH model: it shows that the borrower stands a chance to graduate into a club of countries with low debt and stable consumption and investment.

5.5. Persistent shocks

In the original environment exogenous shocks are i.i.d. Output persistence, being the result of an endogenously fluctuating investment, while significant does not match an empirically observed level. To fix this shortcoming one could introduce exogenous persistence. To this end let the "good" and the "bad" distribution to be state dependent:

$$g(y_j|I,y_k)=\lambda(I)g_1(y_j|y_k)+(1-\lambda(I))g_0(y_j|y_k).$$

By appropriately choosing $g_1(\cdot|y_k),g_0(\cdot|y_k) \quad \forall k$ output could be persistent even if I were constant. One could think of this formulation as the one where output is a sum of two components. First component is a persistent process generated by a publicly observed, potentially stable, level of investment. Second component is a result of an unobserved investment.

As before we allow only two output states and postulate the following transition probabilities[36,37]:

$$\lambda(I)\begin{bmatrix} p & 1-p \\ 0 & 1 \end{bmatrix}+(1-\lambda(I))\begin{bmatrix} 1 & 0 \\ 1-p & p \end{bmatrix}$$

[34] Because invested goods last only one period investment level is smaller than in the data. Thus, this model, despite having production, closely resembles an exchange economy in which $ca_t=Y_t-c_t$. Then there is the following trade-off between matching the consumption and the current account data moments[35]:

$$\rho(ca_t,Y_t)\frac{\sigma(ca_t)}{\sigma(Y_t)}=1-\rho(c_t,Y_t)\sigma(c_t)=1-0.9710\cdot 1.1135=-0.0812. \tag{23}$$

If the model is parameterized to match the moments for consumption and output well it *cannot* match the data on current account. On one hand, to match the significant and negative correlation between current account and output (−0.77) current account's volatility must be counter-factually low. On the other hand, to match the standard deviation of current account ($\sigma(ca)=2.65\sigma(y)$) correlation between the latter and output must be close to zero; again, contradicting the data. Given this trade-off a model's test is to fit the consumption data moments well. This can be achieved by reducing the borrower's discount factor by a notch from 0.98 to 0.97. Moments for this alternative specification are presented in Table 1 in the online appendix.

[35] Using the definition of correlation we get $\rho(ca_t,Y_t)=[\sigma(Y_t)-\rho(c_t,Y_t)\sigma(c_t)]/\sigma(ca_t)$.

[36] For the above formulation to be useful, however, we must restrict the number of states. If we allowed a large number of states, distributions $g_1(\cdot|y_k),g_0(\cdot|y_k)$ could be "hand-crafted" to match potentially any evidence.

[37] Equivalent representation is

$$g(y_2|I,y_1)=\lambda(I)(1-p), \quad g(y_2|I,y_2)=\lambda(I)(1-p)+p.$$

This formulation treats both output states symmetrically and implies that, as before, the efficient level of investment is independent of the current output state:

$$I^*(n,y_j) = (\lambda')^{-1}\left(\frac{\theta}{\beta_c(y_2-y_1)(1-p)}\right) \quad \forall n,\ y_j.$$

The i.i.d parametrization, analyzed before, is a special case with $p=0$. Positive levels of p are associated with positive persistence levels. The highest level of persistence is achieved with $\lambda(I)=0.5$ implying $\rho(y_t,y_{t+1})=p$. Correlation above p can be interpreted as endogenously generated persistence.

We now turn to simulation results reported in Table 3. I update the parameters as follows: $\theta=0.102, \nu=0.90, p=0.50$. First, output persistence is increased across all models. Comparing the LE and the MH models, output persistence increases from 0.50 to 0.65. The increase is smaller than in the case with i.i.d shocks. With persistent shocks there is a spread in continuation values, adding to incentives for investment, even if the country's net worth did not vary across states. So, persistence alleviates moral hazard problem. Yet, all moments are improved notably. Interest premia are significant, volatile and counter-cyclical, close to the levels observed in the data. The trade balance, while still pro-cyclical, is significantly less correlated with output than in the LE model. Moral hazard impedes risk-sharing significantly: consumption volatility is close to that of output and the two series are nearly perfectly correlated. The increased consumption volatility and correlation with output are due to the exogenously introduced persistence. With persistent shocks an output increase or decrease leads to a persistent improvement or worsening of the borrower's net worth and a stronger consumption reaction. Investment volatility is closer to the data because moral hazard problem is not as severe and investment is less sensitive to changes in the borrower's net worth. Also the correlation between investment and output increases. In the case of i.i.d. shocks investment was constant for a range of net worth levels (see Fig. 3, panel A). With persistent shocks investment policy now also varies positively with output.

The MH+LE and the MH models are nearly identical to the LE model. Thus, as before, the limited enforcement has very little effect on the model performance. The moral hazard friction, on other hand, improves the model fit along many dimensions as explained above.

5.6. Model testing

In the spirit of Deaton (1991) we can test the model by comparing implications of different frictions for the joint distribution of consumption and income.[38] Suppose $y_{t-1}=y_1$ realized. In both MH and LE economies investment declines in $t-1$ shifting probability mass toward low income realization. After the more likely $y_t=y_1$ outcome consumption declines mildly in the LE economy (insurance) and it plummets in the MH economy (incentives provision). That is even after several adverse income realizations consumption is likely to stay above the average in the LE but not in the MH economy.[39] This downward consumption rigidity in the LE economy leads us to expect the following relation:

$$\text{in the LE economy}: \ \rho(c_t,y_t|y_{t-1}=y_1) < \rho(c_t,y_t|y_{t-1}=y_2),$$
$$\text{in the MH economy}: \ \rho(c_t,y_t|y_{t-1}=y_1) > \rho(c_t,y_t|y_{t-1}=y_2).$$

Table 4 compares the two correlations in the data and in the model with persistent shocks. Standard errors are reported in parentheses. In the data the correlation between current consumption and output conditioned on the low income realization ($y_{t-1}<E(y_t)$) is stronger. In the LE enforcement economy, as expected, the correlation between consumption and output is stronger when conditioned on high income realization. In the MH economy the correlation is marginally higher conditional on the low past income realization. While the fit is imperfect, the MH economy is certainly closer to the data.[40] It is important, however, to remind that this test hinges on the assumption that state-contingent contracting is possible.

6. Conclusions

This work analyzes a model with moral hazard and limited enforcement in a small open exchange economy. I find that when state contingent contracting is allowed adding the moral hazard friction improves the model's fit along several dimensions. First, it justifies why non-contingent debt is an optimal way to finance an emerging economy. Second, it explains the limited consumption risk-sharing and high, volatile and counter-cyclical interest rates. Third, it generates realistic crisis-like dynamics in which capital inflows are brought to a halt and later reversed and interest rates sky-rocket. The model also has a strong internal propagation mechanism.

The limited enforcement friction, alone or together with the moral hazard friction has nearly no effect on the model's performance. Finally, I show that if the limited enforcement were the dominant friction then correlation between

[38] Kinnan (2012) builds upon this insight and derives testable predictions that allow distinguishing moral hazard, hidden income and limited enforcement frictions. Unlike this work, Kinnan considers a model with a planner facing multiple agents.

[39] For sample simulated paths see Fig. 3 in the online appendix.

[40] The statistics for the "i.i.d" economy are $(\rho(c_t,y_t|y_{t-1}=y_1),\rho(c_t,y_t|y_{t-1}=y_2))=(0.76,\ 0.99)$ in the LE economy and $(0.95,\ 0.78)$ in the MH economy. Thus, the MH economy with i.i.d shocks matches the data qualitatively.

Table 4
A MH vs. LE test.

Statistic	Data	LE	MH
$\rho(c_t, y_t \mid y_{t-1} = y_1)$	0.98 (0.004)	0.61	0.97
$\rho(c_t, y_t \mid y_{t-1} = y_2)$	0.91 (0.033)	0.99	0.96

consumption and output should be stronger conditional on high past output realizations. But in the data this correlation is stronger when conditioned on low past output realizations. The moral hazard friction, on the other hand, is largely consistent with the data.

The model's most important shortcoming is inability to explain the current account dynamics. Aguiar and Gopinath (2007) show that the current account facts can be matched well in the model with trend shocks. Introducing persistent shocks into the model presented here improved its performance along this dimension only marginally. It is likely that the model's performance would be further improved if extended to include stochastic growth.

Finally, while investment played a crucial role for the model dynamics, there is no direct data analog to the one-period capital used here. Kehoe and Perri (2002) show, in a general equilibrium model with capital, that limited enforcement actually makes capital flows more pro-cyclical. I believe that it is promising to explore a moral hazard model and multi-period capital.

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Appendix A. Euler equations

To solve optimization problem (10) one has to maximize Lagrangian

$$\mathcal{L}_{Atk} = u(n+b-\theta I) + \beta \sum_{j=1}^{s} g(Y_j|I)V(Y_j-d_j) + \kappa\left(\beta_c \sum_{j=1}^{s} g(Y_j|I)d_j - b\right) + \phi\cdot(M-b)$$
$$+\mu\left(-\theta u'(n+b-\theta I) + \beta\lambda'(I)\sum_{j=1}^{s}\Delta g_j V(Y_j-d_j)\right) + \beta\sum_{j=1}^{s} g(Y_j|I)\gamma_j(V(Y_j-d_j) - V_{\text{aut}}(Y_j))$$

with respect to current controls $(I, b, d_1, \ldots, d_n)$ and minimize with respect to Lagrange multipliers $\kappa, \mu, \gamma, \phi \geq 0$. The first order necessary conditions and the envelope theorem imply

$$V'(n) = u'(n+b-\theta I) - \mu\theta u''(n+b-\theta I) = V'(Y_j-d_j)\left(1+\gamma_j+\mu\frac{\lambda'(I)\Delta g_j}{g(Y_j|I)}\right) + \phi. \tag{A.1}$$

The optimality condition for the model with moral hazard only is

$$V'_{\text{mh}}(n) = V'_{\text{mh}}(Y_j-d_j)\left(1+\mu\frac{\lambda'(I)\Delta g_j}{g(Y_j|I)}\right) + \phi. \tag{A.2}$$

Appendix B. Computational algorithm

Let k denote the iteration number. Fix the borrowing limits $\overline{d}^k = (\overline{d}_1^k, \ldots, \overline{d}_S^k)$. The operator defined by 10 is a contraction. The value function can be, thus, computed iteratively. Once the value function is computed the borrowing limits can be updated: $\overline{d}_j^{k+1} = Y_j - (V^k)^{-1}(V_{\text{aut}}(Y_j)) \quad \forall j$. Starting with $d^0 = \infty$ so-constructed sequence of borrowing limits is monotone. Because the sequence is bounded below by a vector of zeros it must converge. This computational procedure while guaranteed to converge is time inefficient. I speed up the algorithm by updating the borrowing limits with each iteration on the value function using the following updating rule:

$$\overline{d}_j^{k+1} = 0.5\overline{d}_j^k + 0.5[Y_j - (V^k)^{-1}(V_{\text{aut}}(Y_j))].$$

Appendix C. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org.10.1016/j.jmoneco.2012.11.006.

References

Aguiar, M., Gopinath, G., 2006. Defaultable debt interest rates and the current account. Journal of International Economics 69 (1), 64–83.
Aguiar, M., Gopinath, G., 2007. Emerging market business cycles: the cycle is the trend. Journal of Political Economy 115 (1), 69–102.
Alvarez, F., Jermann, U., 2001. Quantitative Asset Pricing Implications of Endogenous Solvency Constraints. Review of Financial Studies 14 (4), 1117–1151, http://dx.doi.org/10.1093/rfs/14.4.1117.
Arellano, C., 2008. Default risk and income fluctuations in emerging economies. The American Economic Review 98 (3), 690–712.
Atkeson, A., 1991. International borrowing with moral hazard and risk of repudiation. Econometrica 59 (4), 1069–1089.
Atkeson, A., Lucas, R., 1992. On efficient distribution with private information. Review of Economic Studies 59 (3), 427–453.
Clementi, G., Cooley, T., DiGiannatale, S., 2010. A theory of firm decline. Review of Economic Dynamics 13, 861–885.
Clementi, G., Hopenhayn, H., 2006. A theory of financing constraints and firm dynamics. Quarterly Journal of Economics 121, 229–265.
Deaton, A., 1991. Saving and liquidity constraints. Econometrica 59 (5), 1221–1248.
Eaton, J., Gersowitz, M., 1981. Debt with potential repudiation: theoretical and empirical analysis. Review of Economic Studies 48 (2), 289–309.
Economist, 2011. Protectionism in Argentina: keep out. The Economist (September 24), 69.
Economist, 2012. Don't lie to me Argentina; official statistics. The Economist (February 25), 18.
Garcia-Cicco, J., Pancrazi, R., Uribe, M., 2010. Real business cycles in emerging countries? American Economic Review 100, 2510–2531.
Gertler, M., Gilchrist, S., Natalucci, F., 2007. External constraints on monetary policy and the financial accelerator. Journal of Money, Credit, and Banking 39 (3), 295–330.
Gertler, M., Rogoff, K., 1990. North-south lending and endogenous domestic capital market inefficiencies. Journal of Monetary Economics 26, 245–266.
Kehoe, P., Perri, F., 2002. International business cycles with endogenous incomplete markets. Econometrica 70 (3), 907–928.
Kinnan, C., 2012. Distinguishing barriers to insurance in thai villages, NWU manuscript.
Mendoza, E., Yue, V., 2012. A General Equilibrium Model of Sovereign Default and Business Cycles. The Quarterly Journal of Economics 127 (2), 889–946, http://dx.doi.org/10.1093/qje/qjs009.
Perri, F., Neumeyer, A., 2005. Business cycles in emerging economies: the role of interest rates. Journal of Monetary Economics 52 (2), 345–380.
Rogerson, W., 1985. The first-order approach to principal–agent problems. Econometrica 53 (1), 1357–1368.
Rogoff, K., 2002. Moral hazard in IMF loans: how big a concern?. Finance and Development 39 (3).
Thomas, J., Worrall, T., 1990. Income fluctuation and asymmetric information: an example of a repeated principal–agent problem. Journal of Economic Theory 51 (2), 367–390.
Yue, V., 2010. Sovereign Default and Debt Renegotiation. Journal of International Economics 80 (2), 176–187.